

Parameter	Description
Acceleration	Time that line speed takes to reach the maximum line speed. Lower speeds take proportionally less time.
Deceleration	Time that the line speed takes to reach zero speed and standstill. Lower speeds take proportionally less time.

Table 4.9-8 Winding screen explanations

Deposit

The deposit tab shows the deposit settings for the recipe. The deposit settings are split into types. Select a setting type to show its parameters:

- Deposit 1
- Deposit 2

Deposit: Deposit 1

Figure 4.9-18 RECIPES HMI screen - DEPOSIT 1

Parameter	Description
Product Target (OD)	The deposit target of the finished product, e.g. the OD of an aluminium or Alox layer.
Pre-Gas Target (OD)	Used at the start of a run, before the process is active.
Web Transmission (%T)	In the case where the base substrate is not 100% clear (light transmittance), this parameter allows for an offset to be taken into account. If the user believes that the base substrate is not 100% clear, it should be placed in the inline monitor system, and measured. An offset value will be given which should be then placed in this location. Factory default is 99.9%T
Deposit High Alarm (%)	Sets the high alarm level, which will be indicated to the user by colours on the roll reports and while the machine is operational through the Hawkeye reporting monitor.
Deposit High Warning (%)	Sets the high warning level, which will be indicated to the user by colours on the roll reports and while the machine is operational through the Hawkeye reporting monitor.

Table 4.9-9 Deposit 1 screen explanations

Parameter	Description
Deposit Low Warning (%)	Sets the low warning level, which will be indicated to the user by colours on the roll reports and while the machine is operational through the Hawkeye reporting monitor.
Deposit Low Alarm (%)	Sets the low alarm level, which will be indicated to the user by colours on the roll reports and while the machine is operational through the Hawkeye reporting monitor.
Offsetting ON/OFF	Selects this function on or Off. Offsetting checks for and removes errors in the OD of the uncoated web by measuring a short length of it during the chamber evacuation. If the error is within the limits it is automatically corrected. Larger errors prompt the operator to confirm whether to proceed or not.
Offset Low Limit (%)	The low level is set here
Offset High Limit (%)	The high level is set here

Table 4.9-9 Deposit 1 screen explanations

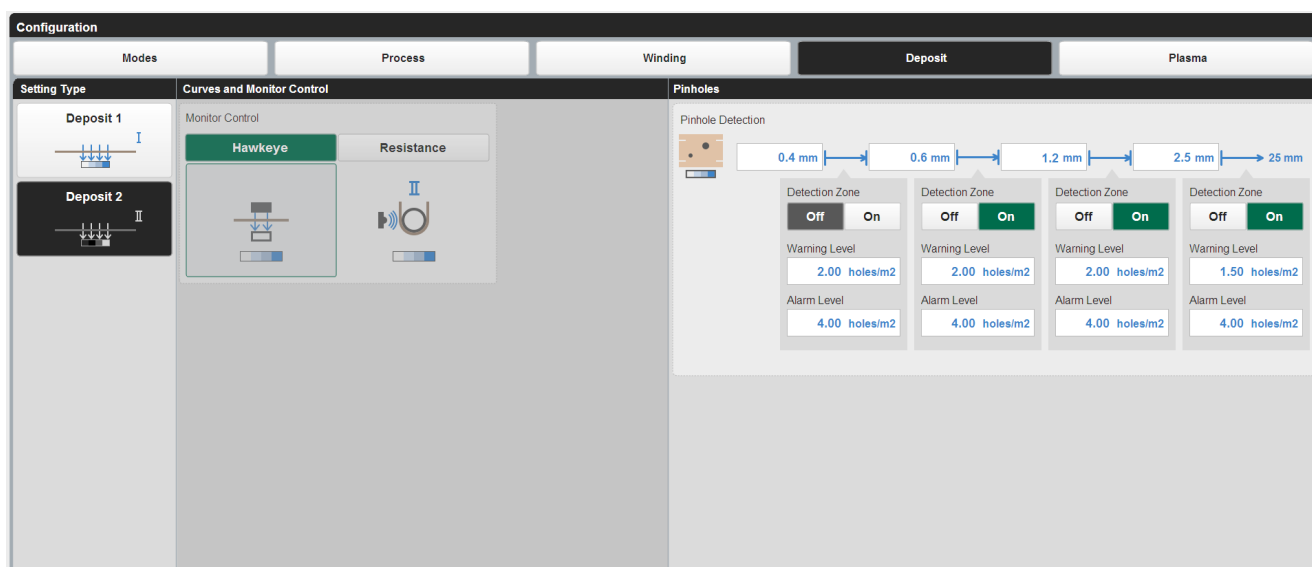
Deposit: Deposit 2

Figure 4.9-19 RECIPES HMI screen - DEPOSIT 2

Parameter	Description
Detection Holes size (mm)	Allows the selection of the size of the holes to be detected over the four ranges
Pinhole Detection Zone On/Off	Switches the detection on or off for the range being selected
Warning Level (holes/m2)	Allows the Hawkeye reporting display and reporting software to make the user when a certain number of holes have been detected
OD Curve Selection	Allows the user to select one of the stored OD curves
OD Curve View	Allows the user to view the selected curve for checking purposes

Table 4.9-10 Deposit 2 screen explanations

Parameter	Description
Monitor Control Type	Allows the user to select which monitor is used for controlling the deposition if more than one is supplied with the machine

Table 4.9-10 Deposit 2 screen explanations

Plasma

The deposit tab shows the plasma settings for the recipe.

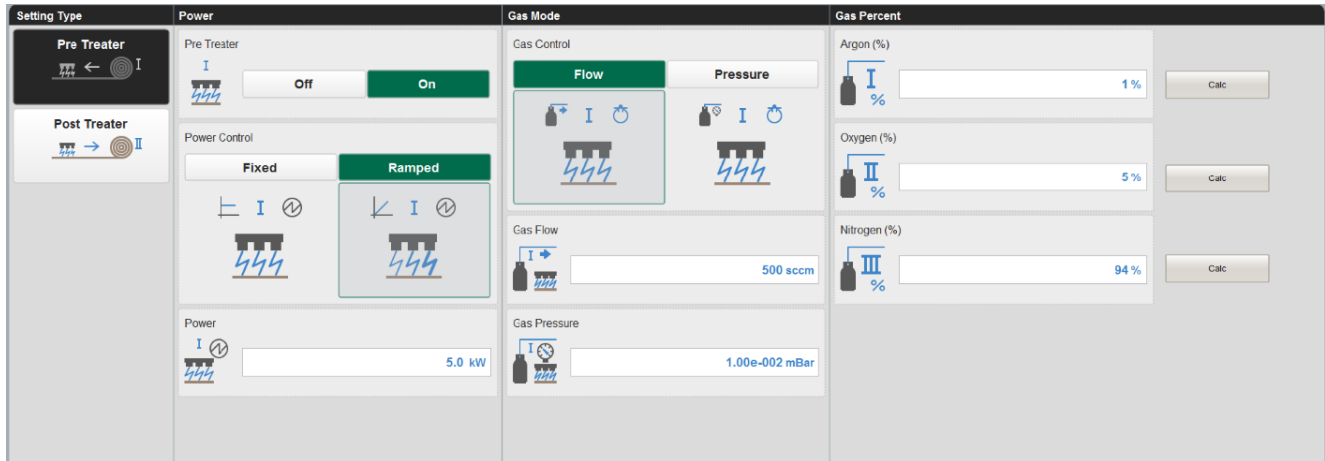


Figure 4.9-20 RECIPES HMI screen - PLASMA – PRE-TREATER

Parameter	Description
Power On/Off	Selects if the treater is to be switched ON or OFF
Power Control mode	Allows the mode of operation to be set.
Power Set point	Allows the input of the required power set point to be applied
Gas Control Mode	Allows the method of control for the process gas control delivery
Gas Flow Setpoint (sccm)	Allows the set point to be inputted, which will be used when the gas flow mode is selected
Gas Pressure Set point (mBar)	Allows the set point to be inputted, which will be used when the pressure mode is selected
Argon Gas (%)	Allows the user to select the amount of the overall process gas to be delivered to the system, as a percentage of 100%, in combination with other process gases
Oxygen (%)	Allows the user to select the amount of the overall process gas to be delivered to the system, as a percentage of 100%, in combination with other process gases
Nitrogen (%)	Allows the user to select the amount of the overall process gas to be delivered to the system, as a percentage of 100%, in combination with other process gases

Table 4.9-11 Plasma pre-treater screen explanations

Loading of Recipe

Once the user has entered the relevant information, the recipe can then be downloaded to the control program by pressing the LOAD TO MACHINE button. The parameters are checked, and any problems are shown in a pop-up.

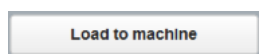


Figure 4.9-21 RECIPES HMI screen - Recipe Load Button

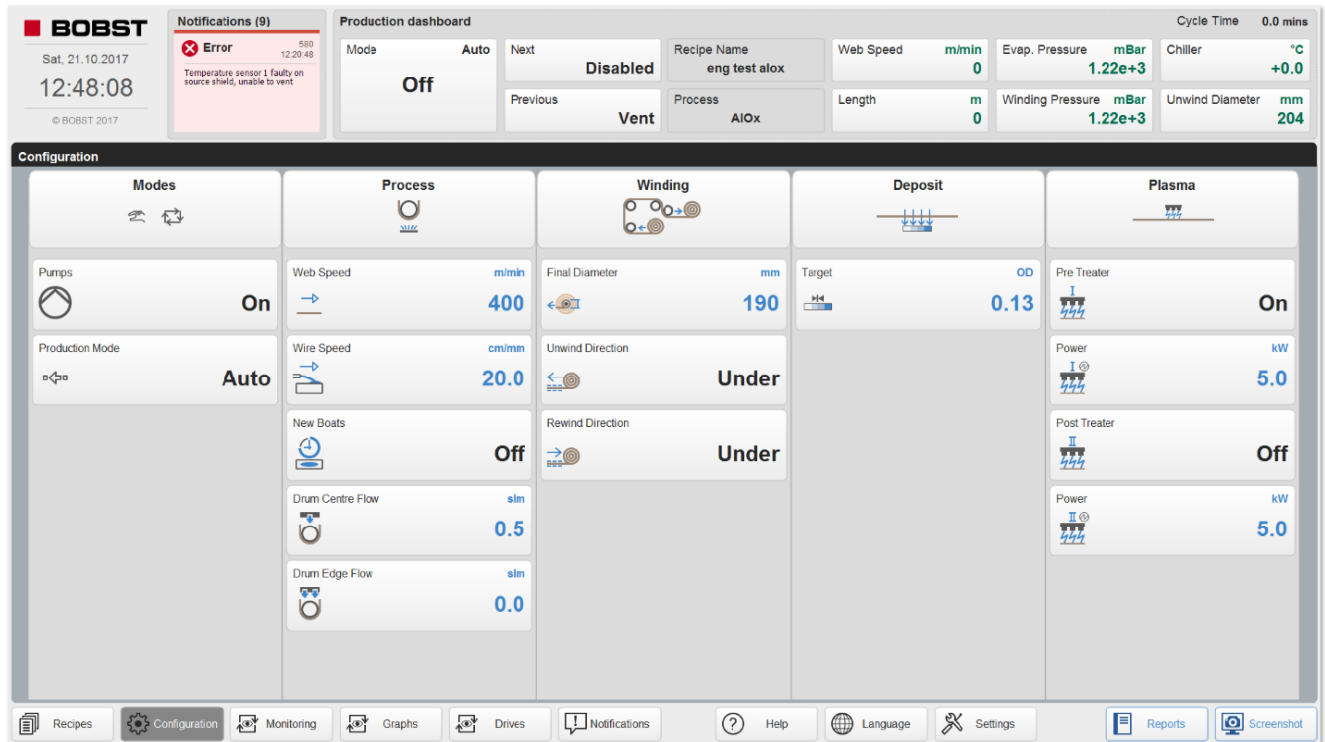


Figure 4.9-22 CONFIGURATION HMI screen

Save Button (Higher Level Access Logon)

To allow the user to make changes during a process cycle and then save these changes, a SAVE LIVE function button has been provided.

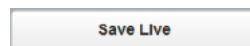


Figure 4.9-23 RECIPES HMI screen - SAVE LIVE button

Press the button. A pop up screen will be shown confirming that the user wants to save the recipe, with the same name.

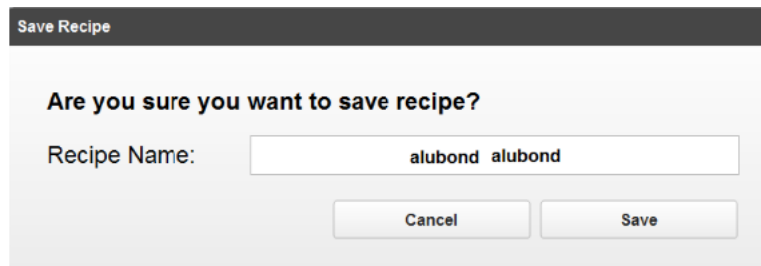


Figure 4.9-24 SAVE Confirmation message

Use the existing name, or enter a new one by touching the data entry area and using the keyboard. Press SAVE to complete.

Deleting RECIPE Files (Higher Level Access Logon)

To remove an existing recipe file, touch the recipe on the FAVORITES or ALL RECIPES HMI screen and press the DELETE button.



Figure 4.9-25 DELETE button

A further pop up window should will be shown, asking the user to confirm the recipe file by name.

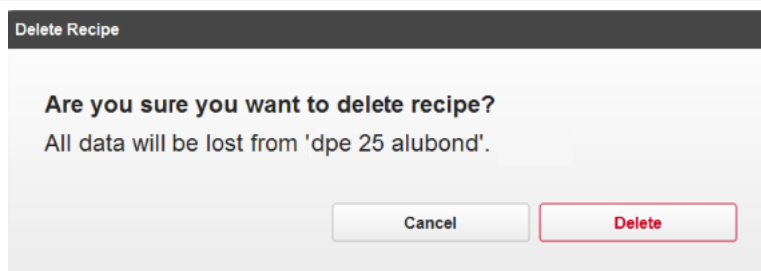


Figure 4.9-26 DELETE confirmation pop up screen

In the case, where the recipe has been created by Bobst, the following message will be shown.



Figure 4.9-27 Error message displayed when trying to delete a Bobst created recipe

Possible Displayed Messages

There are a number of possible messages that may be displayed while working in this area of the HMI software.

[See chapter 9.3 HMI Status Messages on page 415 for details.](#)

4.10 End of Roll Sensing (Auto Slowdown)

A function of the substrate winding drive system is to detect when either of the unwind or rewind rolls are reaching their minimum or maximum diameters and then stop the drive system from rotating.

The function is designed to trigger the deceleration of the drive line speed to zero and to stop if the unwind is running off the core, or the rewind exceeding the maximum specified roll diameter parameter.

Normally it is initialised by the material on the unwind roll approaching the core. However at the same time, there is a requirement to leave a minimum amount of material on the unwind core.

The machine calculates the unwind roll diameter at which the drive system should start to slow down to finish at the final unwind roll diameter, based upon the following operator inputted information.

Required FINAL diameter of material to be left on the core

The information is entered into the RECIPES OVERVIEW or CONFIGURATION WINDING HMI screen, by the user, and loaded onto the HMI.

The screenshot displays the 'Recipes' HMI interface with four tabs: Overview, Process, Winding, and Deposit. The 'Overview' tab is active. The interface is divided into two main sections: 'General' and 'Web'.

General Section:

- Roll ID:** 21102017-124525
- Product Target:** 0.13 OD
- Wire Speed:** 20.0 cm/min
- Web Speed:** 400 m/min

Web Section:

- Web Width:** 1100 mm
- Adaptive Width:** Off
- Adapt to New Width:** 1000 mm
- Unwind Final Diameter:** 190 mm (highlighted with a red box)
- Adaptive Width Database:** Adaptive Width Disabled

Figure 4.10-1 Unwind section of the RECIPES OVERVIEW HMI screen.

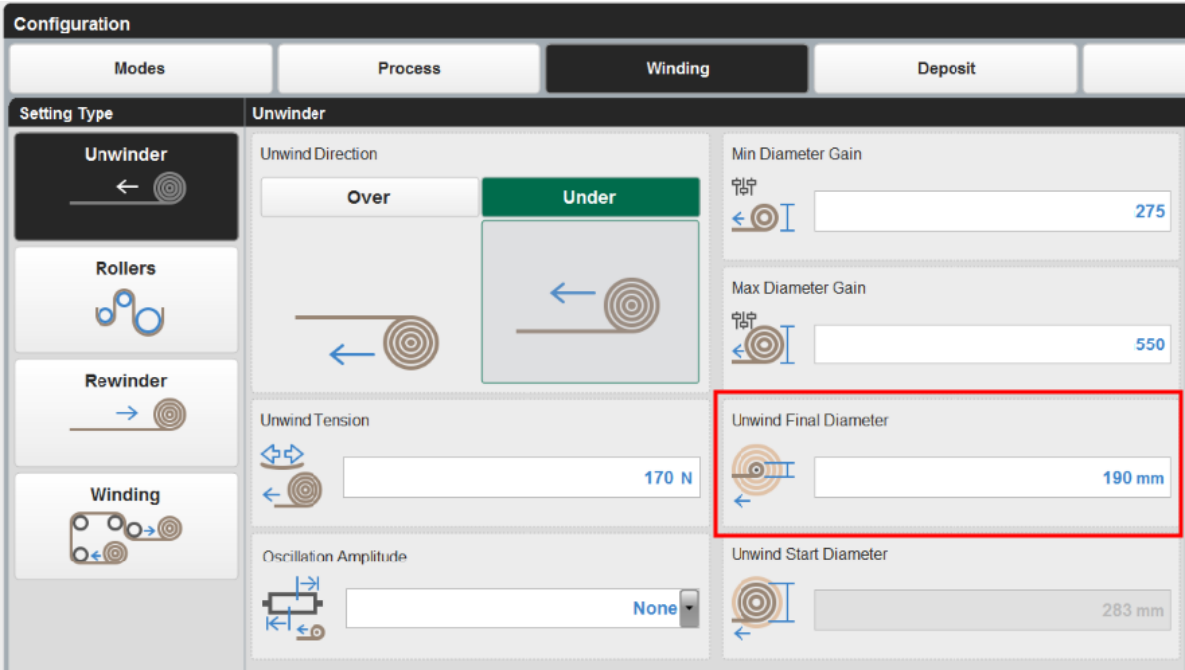


Figure 4.10-2 Unwind section of the CONFIGURATION WINDING HMI screen

The machine will make alterations for speed changes during the coating cycle, automatically, if the operator changes the speed.

When auto slowdown is used, the machine will go through the 'auto slowdown' sequence to vent the chamber up to atmosphere and then wait for the operator to open the machine. [See Figure 7.2-3 on page 325 for details.](#)

4.11 Monitoring Running Conditions

The MONITORING HMI screen will indicate to the user what is occurring with certain components in the drive system at times when the coating process is taking place.

The following list of feedback signals and conditions are monitored:-

- Line speed feedback (m/min)
- Applied unwind tension feedback (N)
- Applied rewind tension feedback (N)
- DTR1 Torque/Draw feedback (N/%)
- DTR2 Torque/Draw feedback (N/%)
- Drum Draw feedback (%)
- Spreader roller apex positions (deg)
- Roller surface speed feedback as a comparison with the line speed (%)

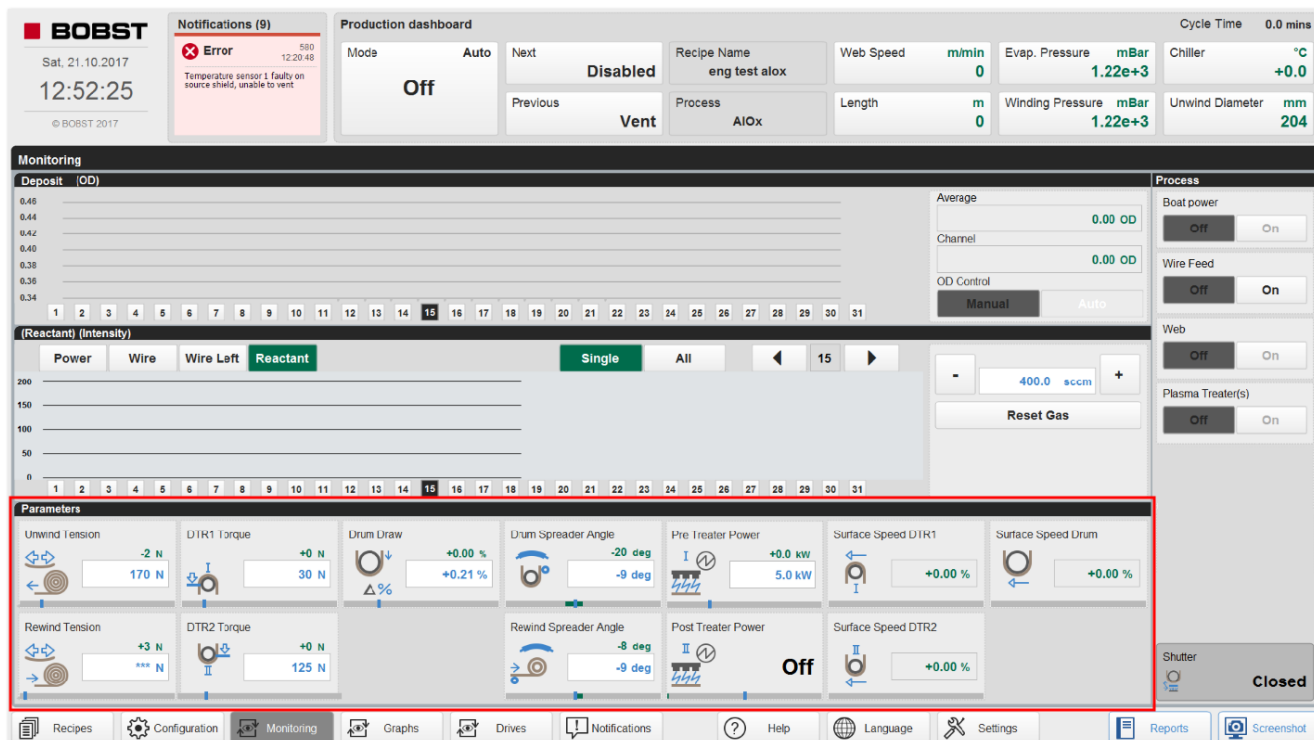


Figure 4.11-1 MONITORING HMI drive components feedback.

For further specific details the user needs to go to the DRIVES HMI screens

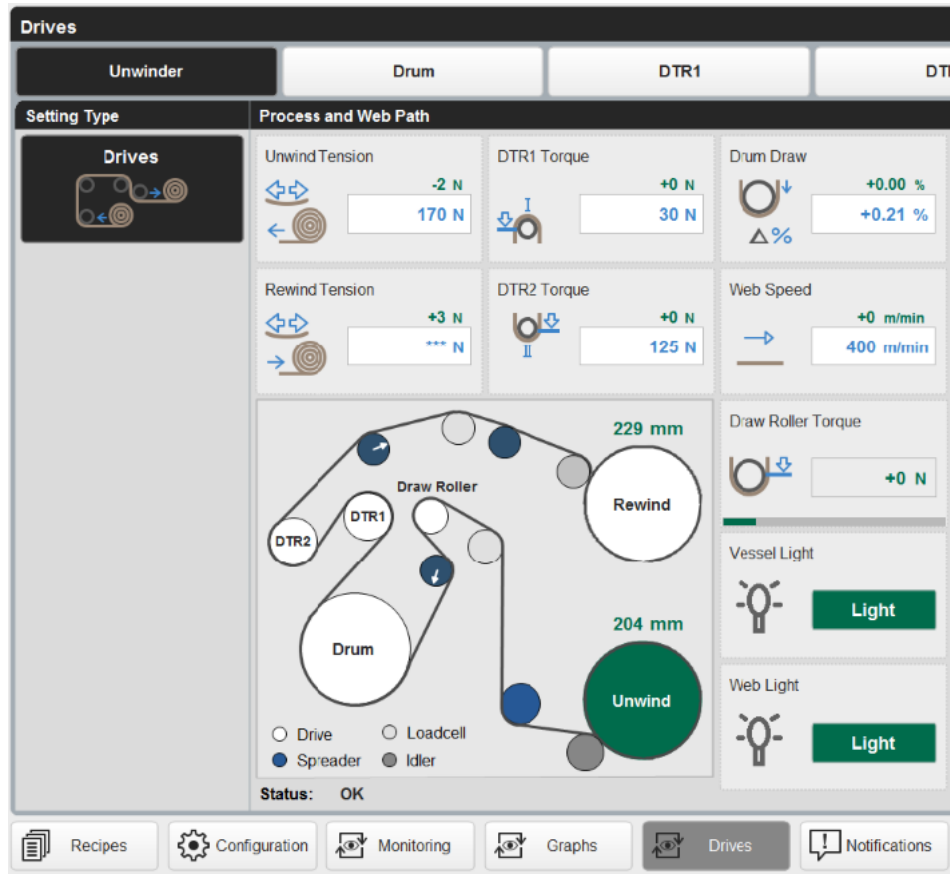


Figure 4.11-2 DRIVES HMI screen

The blue numbers are set pints, while the green numbers are the actual feedback.

Setting/Adjusting Parameters:

The following parameters can be adjusted while the machine is running by entering a number value, between the maximum and minimum range value, in the blue box.

- Line Speed
- Tensions
- Draw Ratio's
- Current/Torque Limits
- Gain Settings

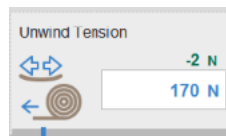


Figure 4.11-3 MONITORING and DRIVES HMI screen adjustments.

Touching the blue parameter, will enable the touch screen number keyboard to be shown. The parameter can be increased or decreased within the minimum and maximum values.

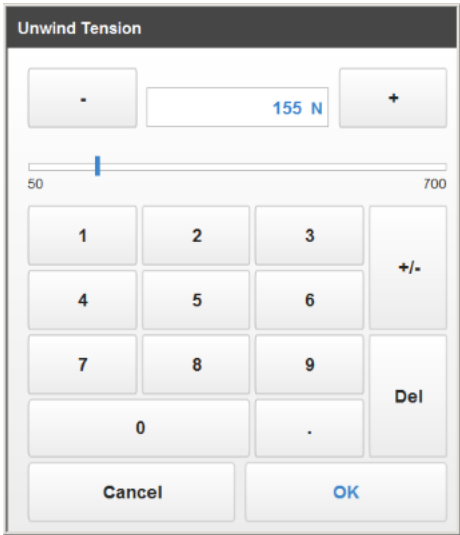


Figure 4.11-4 MONITORING and DRIVES HMI screen - Unwind tension parameter set point modification.

Monitoring Set-point/Actual Operating Parameters

The horizontal bar chart, at the base of the window shown, gives an indication of where the set point is within the full range of adjustment.

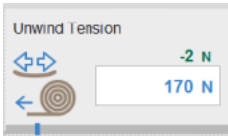


Figure 4.11-5 MONITORING and DRIVES HMI screen feedback.

- Line Speed
- Tensions
- Draw Ratio's
- Current/Torque Limits
- Gain Settings

The blue numbers are set points, while the green numbers are the actual feedback.

Monitoring Unwind Diameter

The unwind diameter is shown on the production dashboard on all HMI screens.



Figure 4.11-6 Production Dashboard section of the HMI screens giving the unwind diameter feedback

The unwind section of the screen allows the user to monitor the performance of the unwind roll's diameter during the coating cycle.

4.12 Gain Settings - Rewind/Unwind

The drive gain setting value can be entered in the unwind or rewind section of the RECIPES- WINDING screen to allow the user a certain degree of control of the response of this drive unit.

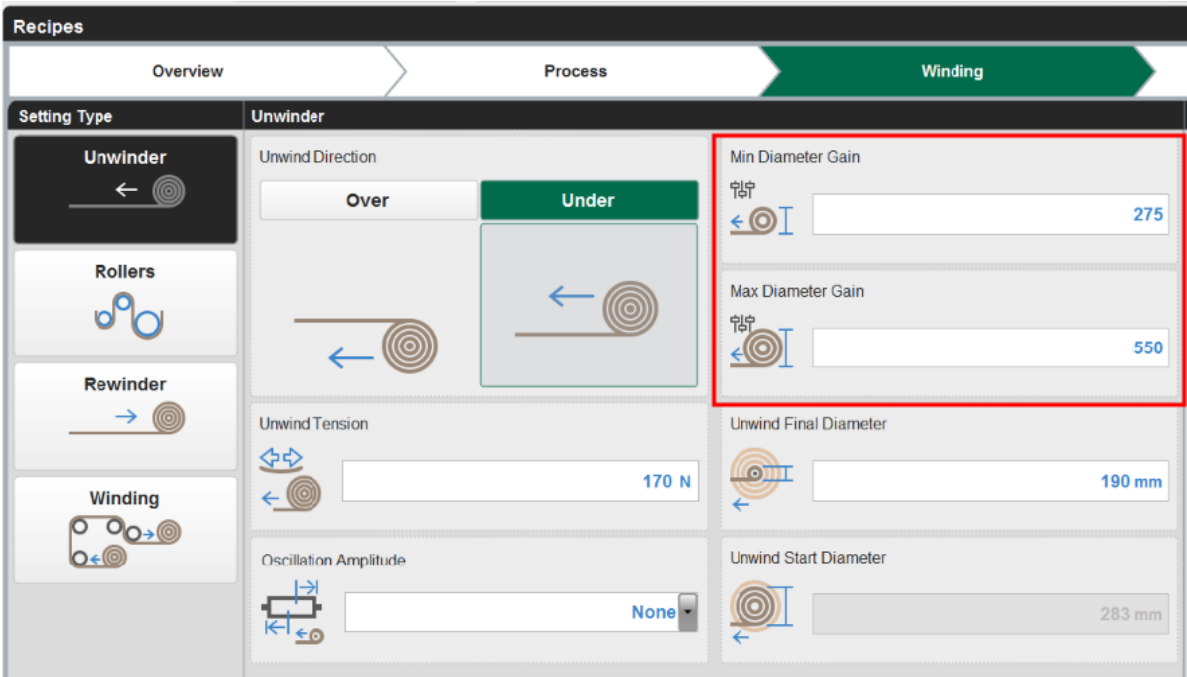


Figure 4.12-1 Gain settings section of SETUP HMI screen - Unwind

Within the computer, there is a function that interpolates between the two gain values vs. the diameter.

There are two values, one for the gain setting at minimum diameter (core) and one value for the maximum roll diameter. The higher the number that is entered into the DRIVE GAIN value, the more responsive the drives will become.

	Note !
	Do not set the levels to a value where the drive becomes unstable.

Once the recipe has been downloaded, the parameters can then be changed in the CONFIGURATION – WINDING screen and through the SETTINGS- DIAGNOSTICS – DRIVES screen.

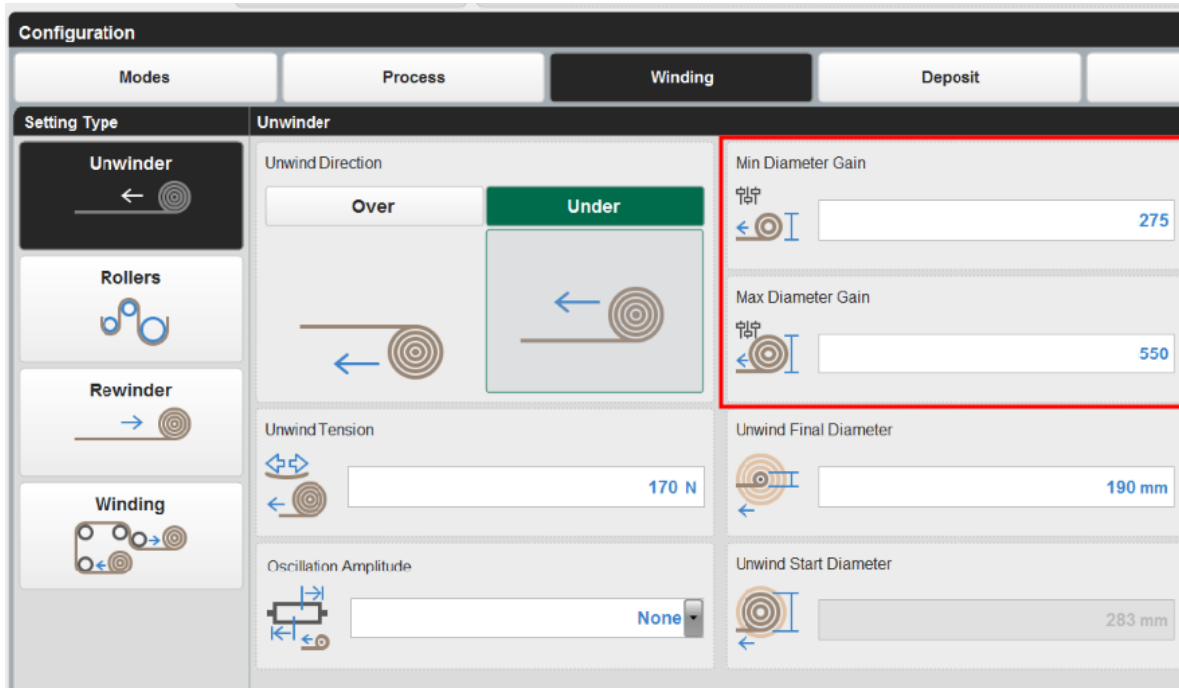


Figure 4.12-2 Gain settings section of CONFIGURATION- WINDING HMI screen - Unwinder

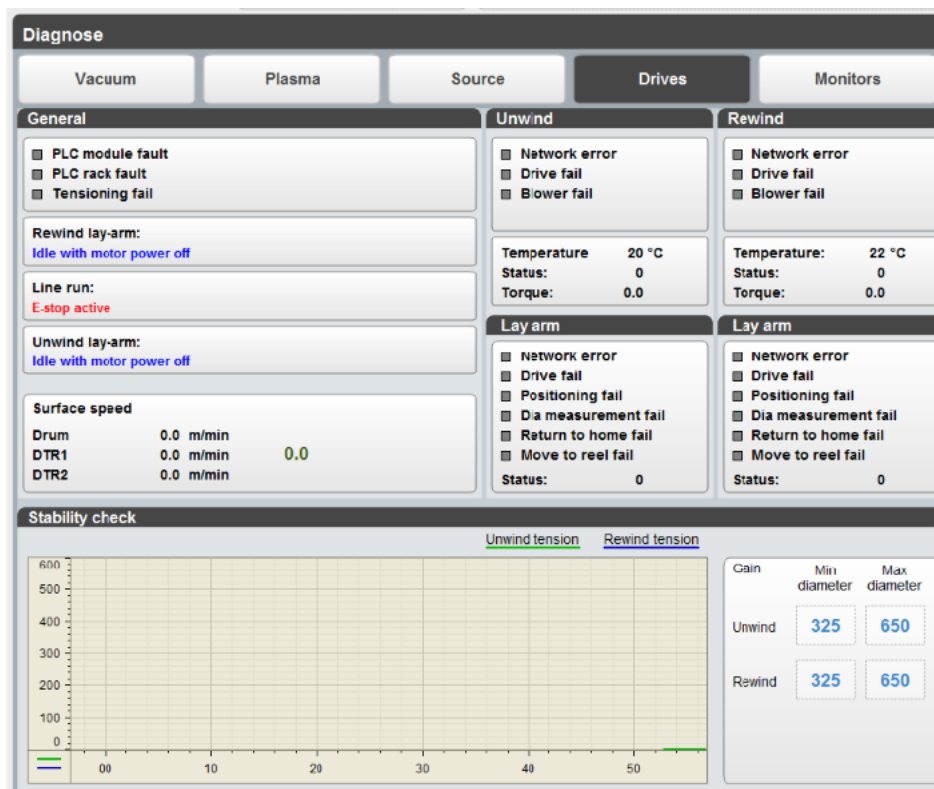


Figure 4.12-3 Gain settings section of SETTINGS- DIAGNOSTICS – DRIVES HMI screen - Unwinder

Generally, higher values should be used for materials such as CPP, LPDE, and smaller values for BOPP, PET and Paper.

Monitoring for responses.

The historical display available from the DIAGNOSE-DRIVES HMI screen can be used to monitor the stability/performance of the gain settings. Selecting the DIAGNOSE-DRIVES HMI screen will allow the user access.

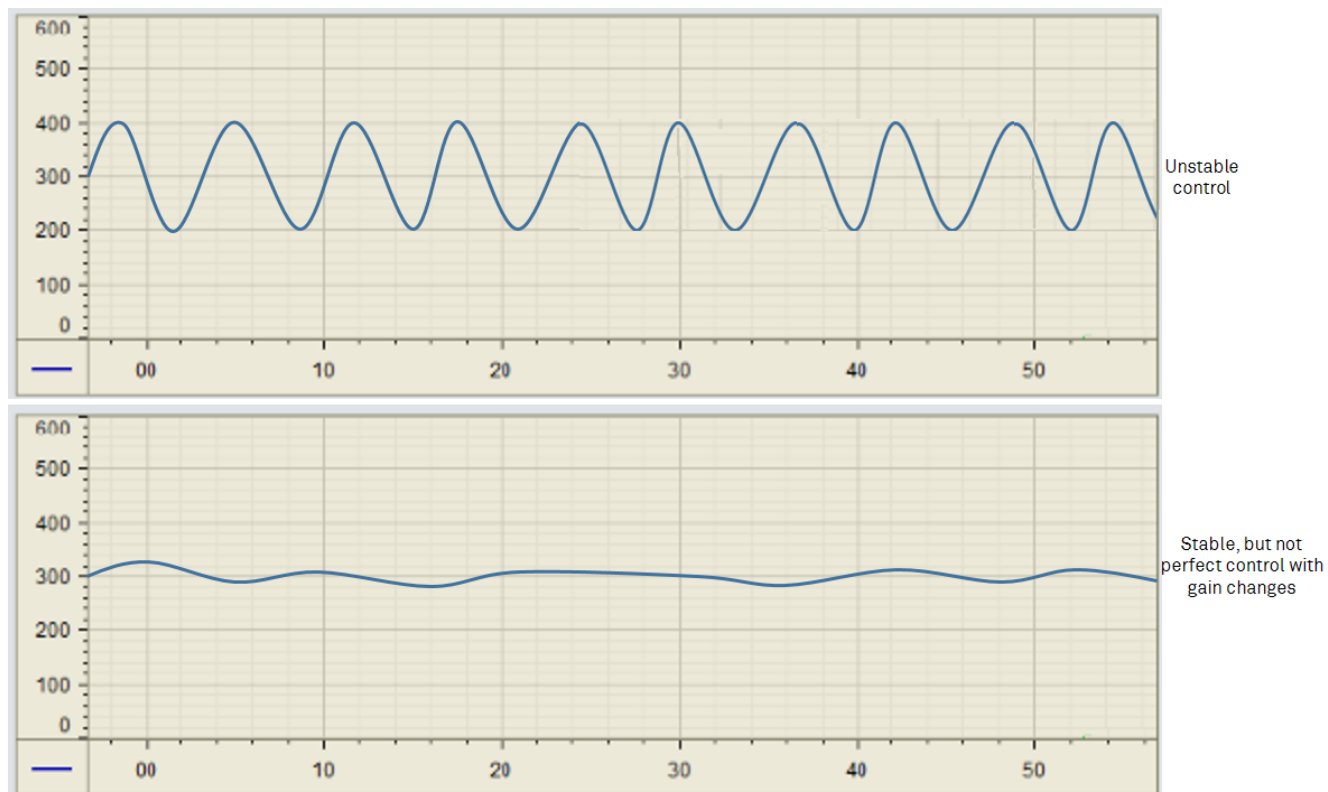
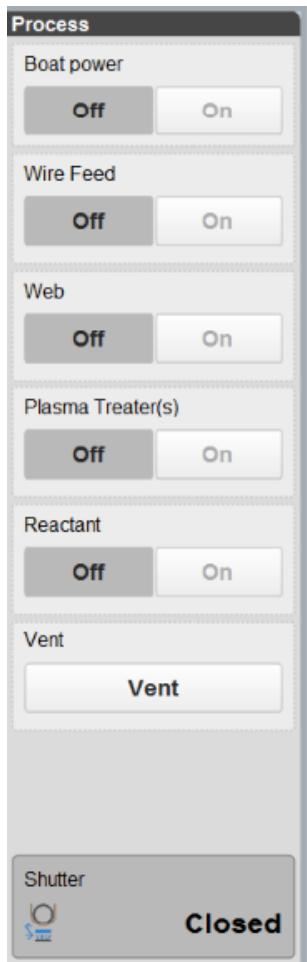


Figure 4.12-4 DIAGNOSE-DRIVES HMI screen Stability Checking

4.13 Starting the Drive System & Selecting Modes of Operation

The drive system has a number of programmed modes of operation that are selected by the various buttons on the DRIVES HMI screen or the local HMI station in the loading area of the machine.

Here is a description for the following functions that are active on the Process HMI screen.



Web On/Off –starts to move web at the WEB SPEED if all the relevant interlocks are made.

Table 4.13-1 MONITORING -PROCESS HMI screen control mode buttons

Here is a description for the following relative function that are active on the rear HMI.



Figure 4.13-1 Rear local HMI station

Table 4.13-2 Rear local HMI controls

- Web Start/Web Stop
- Tension On/Off – applies tension to the web (substrate)
- Reset Drives : Following a drive fault or machine power up
- Inch Drive System :- Moves the drive system at a slow speed in the direction chosen
- Reset Lay-arms (Only appears when arm(s) need to be reset. For example on the machine power up.
- Get Diameter:- Sends the layarms to contact the roll/core

The drive start function is not available unless the winding cart is driven fully into the chamber and tension mode has been activated. When the buttons are activated, the small square icon will move/ highlight within the button.

4.14 Web Break Detection

If there is a rip/break in the substrate, the machine has been designed to stop the drive system automatically.

If not, the coating drum will potentially become coated with aluminium and there is a possibility of a “wrap around” the rollers by the substrate can occur.

The drive system will perform a "Fast Stop" procedure when it detects a web break.

To determine if a web break has occurred, the drive systems monitor both the rewind and unwind tension feedback levels.

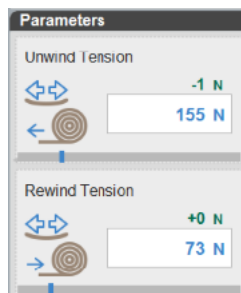


Figure 4.14-1 MONITORING - PROCESS HMI showing tension set point and feedback levels.

If one of these levels falls below a pre-set low level (normally 3% full tension range), it triggers the web break detect circuit.

The drive system will de-accelerate in a pre-designed time between 10 and 20 seconds to a stop condition

After this time period, the drive system is powered down.

“Web break” message will be displayed on the HMI interface.

To reset this condition, the “Drive Reset” button on the Rear HMI interface has to be touched.

Secondary Actions

On a web break condition being sensed, the aluminium wire and evaporators will be powered down automatically, and the shutter will close.

4.15 E-Stop and G-Stop Functions

E-Stop is a short name for emergency stop. G-Stop is short for guard stop.

E-Stop

In an emergency situation the operator should press one of the E-Stop push-buttons on the machine

This will immediately remove the power to the drive system and other rotating equipment. At the same time, all other power circuits will be de-energised. The drive rollers and rolls will coast to a standstill condition if they are still rotating.

The Notifications section of the HMI screen will indicate when an E-Stop is in place. This will be also registered on the Alarm HMI screen.

Resetting the machine after an E-Stop function has occurred

	Warning ! Ensure that all sections and areas of the machine are clear of personnel, before resetting the machine systems.
---	---

After ensuring that the machine is safe to be reset after an E-Stop condition has occurred, the circuit can be reset. The E-Stop Reset button will be flashing after an E-Stop has been performed. This is located on the operator panel (E0_5). Press this button to reset the circuit.



Figure 4.15-1 E05 Panel Reset Estop button.

G-Stop

In normal operation of the machine, there are situations when the traverse isolator is switched to the OFF position. This located on the E06_3 panel.

When activated, the power to the drive system motors and traverse drive will be removed, It also removes pneumatic energy to the shutter.

The Notifications section of the HMI screen will indicate that a G-Stop is in place.

Resetting the machine after a G-Stop function has been activated

	Warning ! Ensure that the winding mechanism and the chamber areas of the machine are clear of personnel, before resetting
---	---

After ensuring that the machine is safe to be reset after an G-Stop condition has occurred, the circuit can be reset. The G-Stop Reset button will be flashing after a G-Stop has been performed. This is located on the operator panel (E0_5). Press this button to reset the circuit.

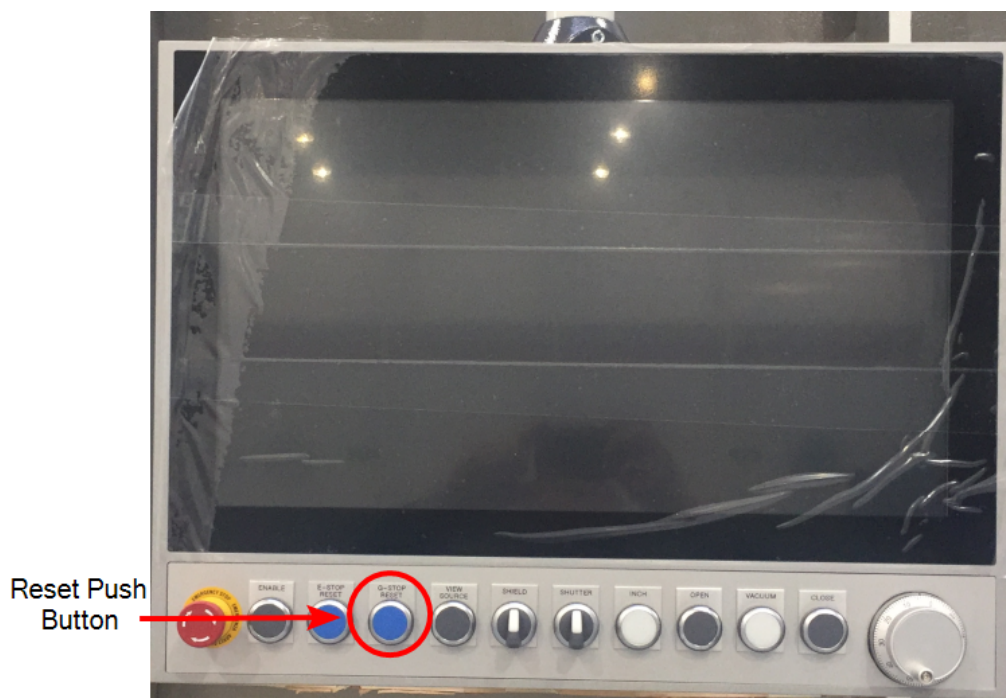


Figure 4.15-2 E05 Panel Reset Estop button.

Drive Reset Function

The Reset Drives button on the Rear HMI screen has to be touched to allow normal operation of the drive system. This will place the drive system in a 'ready state' if all alarm conditions have been overcome.

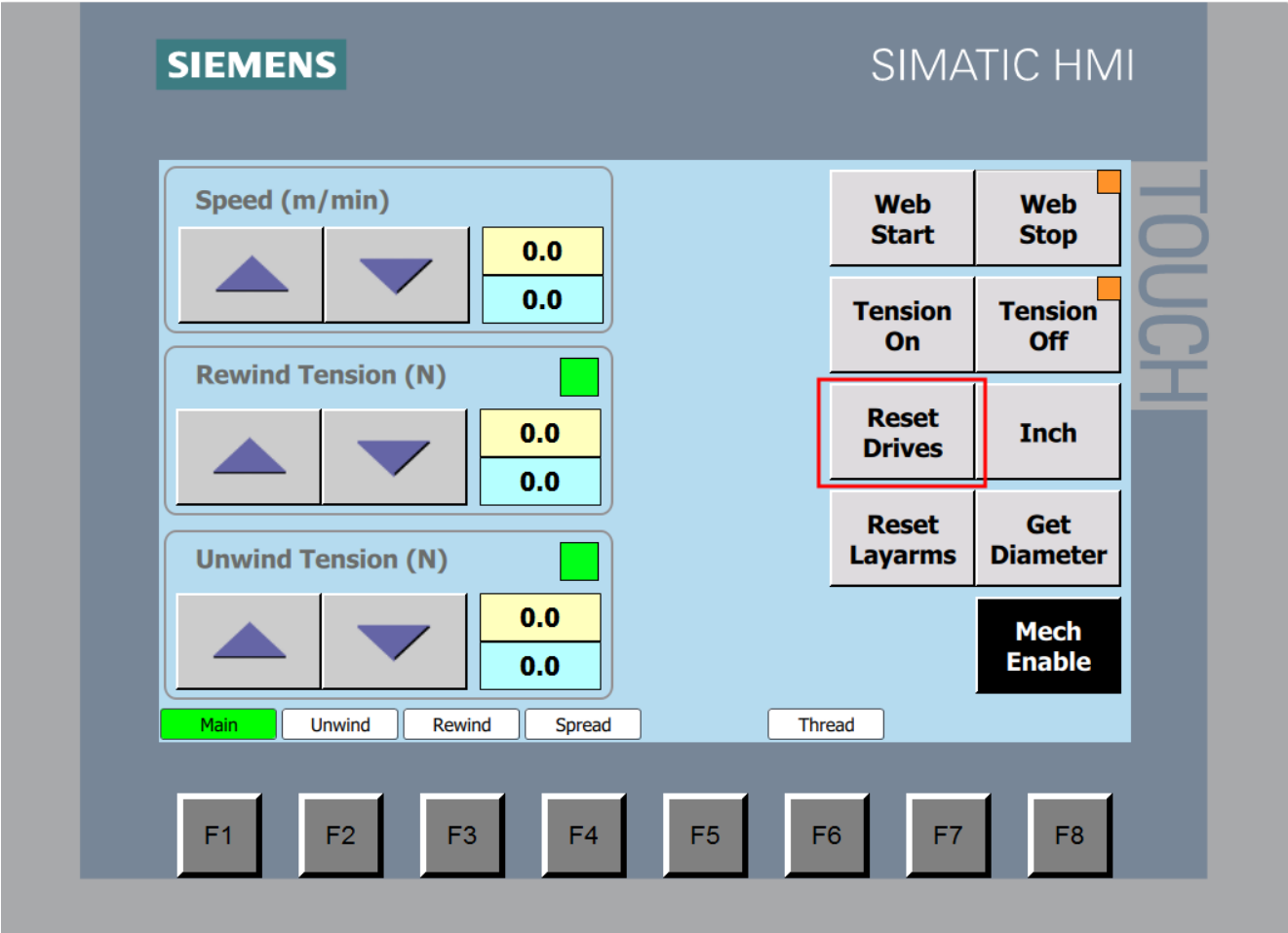


Figure 4.15-3 Example of the reset drives button on the rear HMI screen.

4.16 Web Drive System Inch Function

The web movement drive system can be inched (jogged) slowly by pressing the Inch button on the rear HMI station.



Figure 4.16-1 Rear local HMI station.

Direction of movement

The normal direction of rotation is in the process winding direction.

Roller movement

	Note !
	Before the rollers will move, there is a 5 second delay while the warning lights flash and audible siren operate. These warning indicators will stay activated during this operation.

To comply with the recommended safety requirements (PrEN 1010:1993), the control system is programmed to allow the following sequence;

	Warning !
	Ensure that all relevant sections and areas of the machine are clear of personnel, before continuing with this operation.

- After initial holding down of the push button for a minimum time period of 3 seconds, release the push-button.
- Wait for one second and then press the push button again.
- The rollers will turn if tension is applied to the substrate. If tension is not applied, only the rewind shaft will rotate.
- After the push button has been disabled, the siren and warning lights will be activated for a further 5 seconds.

Inching with the chamber closed

Using the inch button with the chamber closed will rotate the drive system without any audible or visual alarms, as the movement takes place inside the chamber.

4.17 Substrate Winding System Control Principles

The substrate winding system uses industry standard methods of control. These can be classified as “open loop” and “closed loop” solutions. BOBST produce machines that use these two principles to provide a very accurate winding system. This includes speed feedback from motors and the use of load cell systems for measuring the tension applied to the substrate. This information is displayed on the MONITORING HMI screen.

Roller Speed Feedback

The HMI indicates the actual roller surface speed in metres/min.

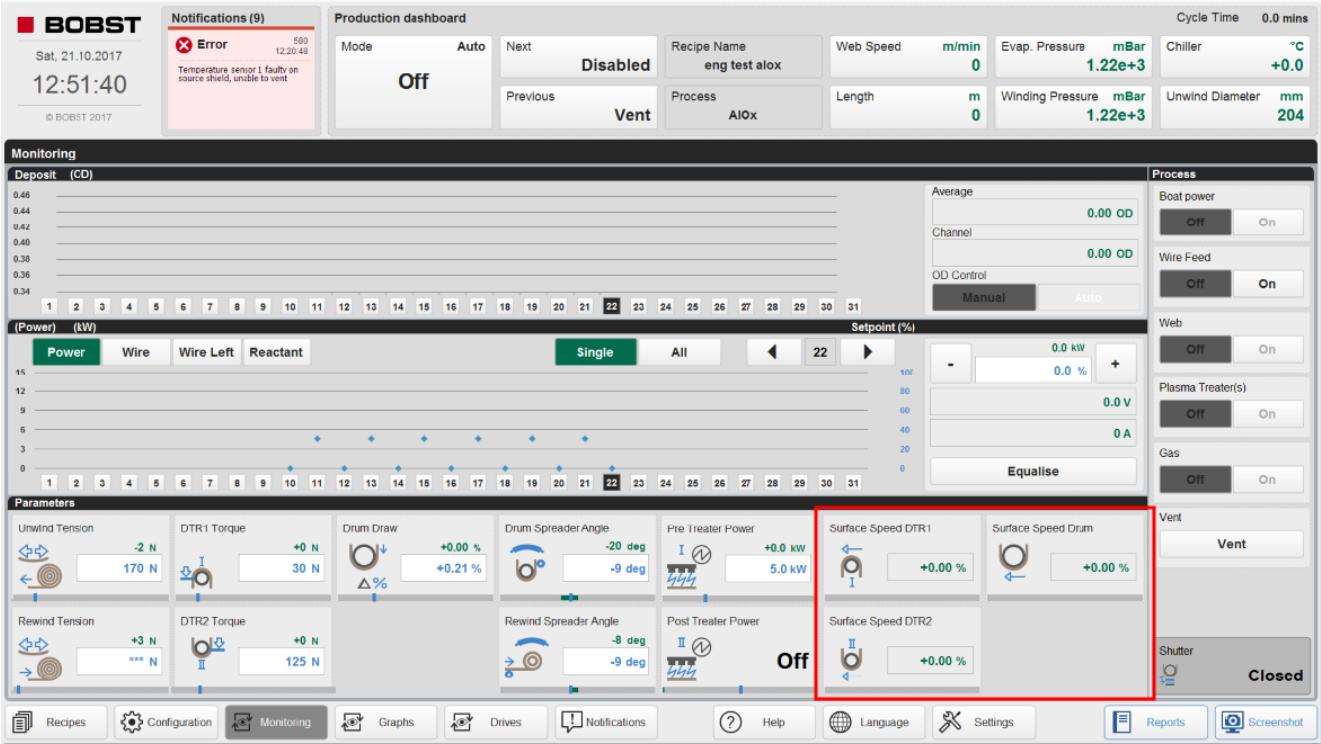


Figure 4.17-1 Roller speed feedback.

This allows the user to understand if slippage is taking place between the substrate and the rollers and allows direct feedback to this situation.

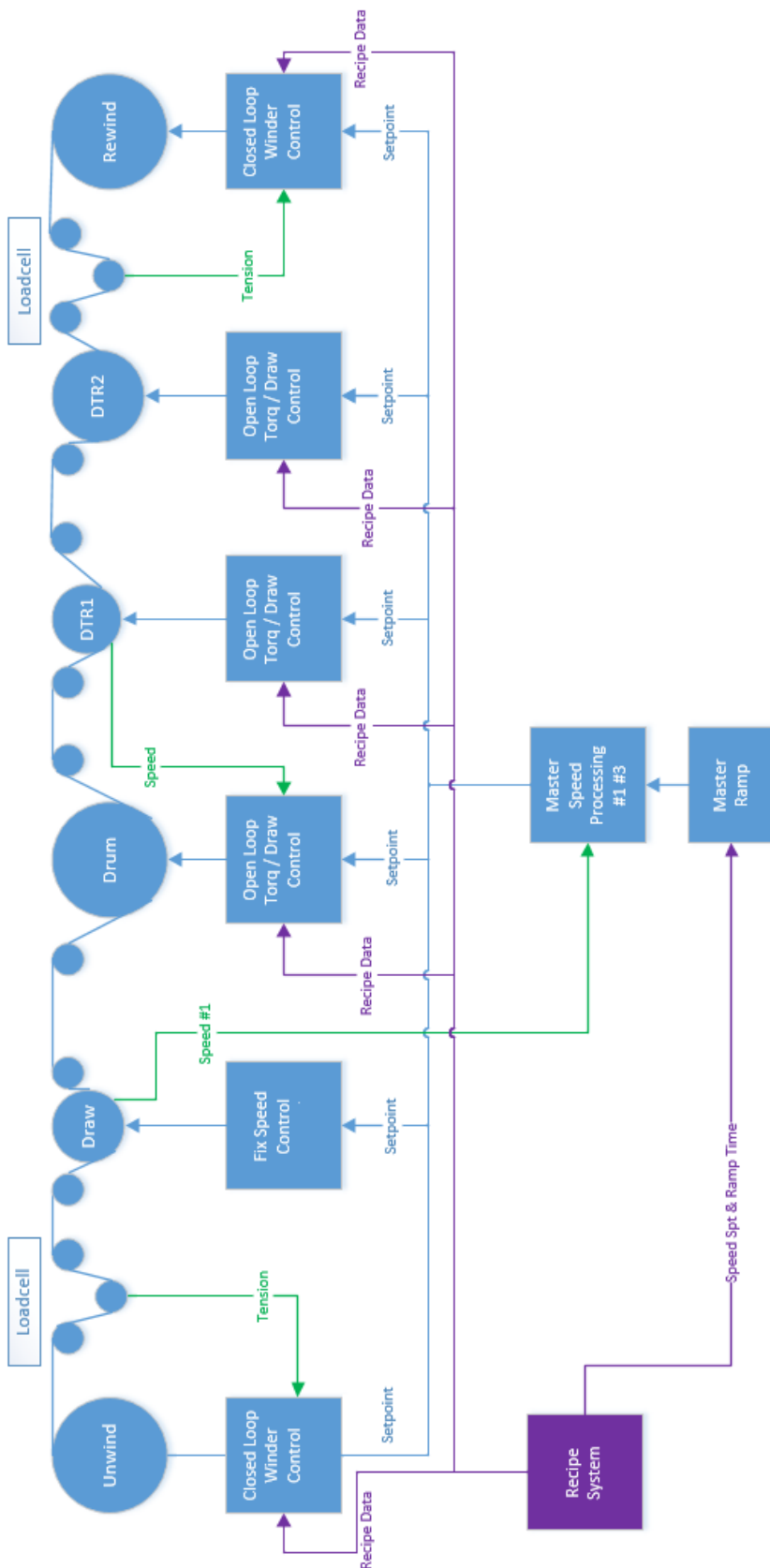


Note !

The winder speeds are not displayed because the diameter is constantly changing during the process, and any value would only be approximate.

The tension feedback values are an indication of the performance of the winders.

The “Line Speed” feedback value is the surface speed of the “DRAW” roller.



Process Driven Rollers

There are different types of process driven rollers.

Draw Roller

The draw roller is the master roller. The CVD1 draw roller is the master roller. It is rotated by a direct motor/drive combination that allows the cork covered roller to be used as part of the winding mechanism. This combination controls the rollers surface speed to equal the required line speed set-point.

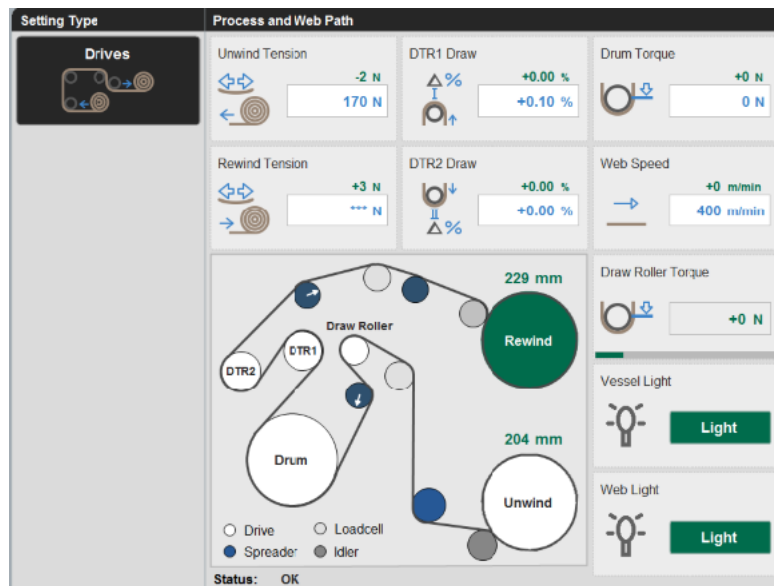


Figure 4.17-2 DRIVES HMI screen – line speed feedback section.

The value is reported to other HMI screens in the production dashboard.

Cooled Process Drum

A process drum is provided to allow the removal of heat from the substrate as it passes through the deposition source zone.

The heat is generated from:

- the aluminium vapour that is condensing upon the substrate
- the evaporators that produce the aluminium vapour (radiant heat)

It is designed and manufactured to ensure an equal cooling performance across its full width.

A cooling liquid is supplied to the process drum from the process chiller at a pre-set temperature set point.

The surface of the drum is manufactured to a fine tolerance. Care should be taken not to damage the drum surface, as this will cause potential problems with overheating of the substrate.

Also, refer to the section of the manual relating to the gas wedge system and its operation.

The calculated surface speed is indicated on the HMI screen. This calculation is based upon the roller diameter and the speed feedback from the motor encoder.

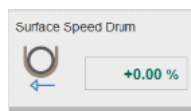


Figure 4.17-3 DRIVES HMI screen – Driven roller feedback section.

Cooled Process Roller (DTR1)

A cooled driven roller is provided to allow the removal of heat from the substrate it is contact with. The cooling liquid is supplied to the process drum from the process chiller at a pre-set temperature set point.

The surface of the drum is manufactured to a fine tolerance. Care should be taken not to damage it.

As an additional option, the cooling to this roller can be isolated by the user using a selection button on the SETUP HMI screen.

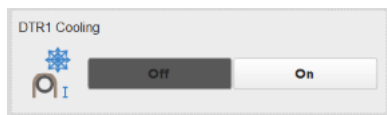


Figure 4.17-4 HMI screen – DTR1 cooling selected/not selected.

If the substrate at the rewind is too cold, and condensation forms on it, in the example of PET, this isolation function can be used.

The calculated roller speed is indicated on the DRIVES HMI screen.

DTR2 Roller

This driven roller is constructed from a non-metallic material called Ebonite. It has a high friction covering to ensure good contact between the substrate and the roller surface. The covering must be cleaned or replaced if the surface is damaged or badly contaminated. This is generally indicated by a DTR2 slipping alarm when the machine is running under normal operating conditions.

The calculated surface speed is indicated on the DRIVE HMI screen.

Description – DTR

The name is an acronym for “drum tensioning roller”. The purpose of these rollers is to apply tension to the substrate, as it passes over the process drum.

Modes of operation

The process drum, DTR1 and DTR2 rollers can be operated in two modes:

- Open Loop – Torque mode
- Closed Loop – Draw mode

Description of Draw Control Function

The Drum, DTR1 and DTR2 rollers are referenced to the DRAW roller speed set point. These are described as controlled rollers.

If the rollers are set to +0.0% in “draw mode”, all the roller speed set points are the same, and the speed feedback should indicate this situation

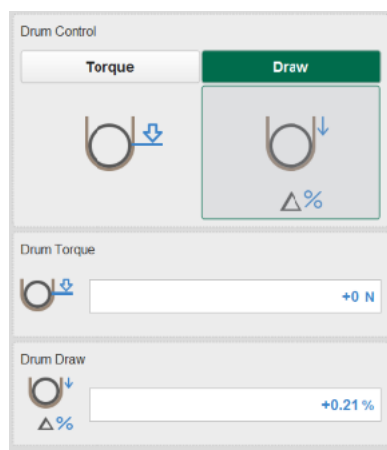


Figure 4.17-5 RECIPES HMI screen roller control mode.

The draw feedback value is described to be zero, when both the reference roller and the controlled roller being monitored are rotating at the same surface speed. The speed of the controlled roller can be adjusted to rotate faster or slower, than the reference roller surface speed, by adjusting the draw set point value.

For example, if the draw value is changed to 0.5%, the controller roller will rotate at a speed 0.5% faster than the DRAW roller surface speed. If the value is set to -0.5%, the controlled roller will rotate at a speed, 0.5% slower than the DRAW roller.

It is usual to operate the process drum and rollers, slightly faster. This allows the substrate to be held tight across the surface of the drum, to provide the best conditions for cooling of the substrate.

This mode of operation is described as a “closed loop system”.

Description of Torque Control Function

An autoloss sequence is carried out as part of the machine start-up procedure. This will determine how much power is required to rotate the rollers under a vacuum condition, to overcome mechanical losses.

This is a calibration task. [See chapter 11.4 Auto-loss Sequence \(Winding Mechanism\) on page 539](#)

	Note !
	The base torque settings for the motor losses are set up using the autoloss (drive losses) sequence under the calibration screen

This mode of operation allows under and over speeding of the controlled rollers (drum, DTR1, DTR2), and the reference DRAW roller. The over speed at low speed is limited to +/- 10% of the line speed, which then changes to +/- 2% over speed at higher speeds. Positive torque settings will create an over speed situation. Negative setting will create a under speed situation

With the torque value set to 0%, the drive motor will not have enough power to over speed the relevant roller.

In fact, the roller either will slip under the substrate or be “pulled” around by the substrate

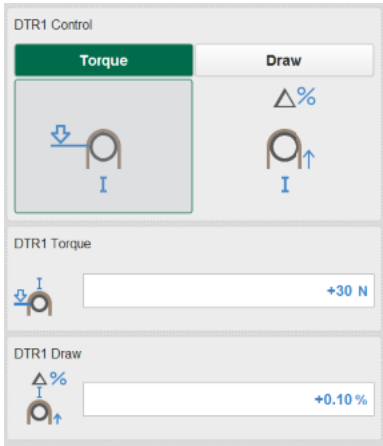


Figure 4.17-6 SETUP HMI screen roller control mode.

The speed of the controlled roller can be adjusted to rotate faster or slower than the DRAW roller, by adjusting the torque value.

For example, if the torque value for the DTR2 roller is set to100 newtons, the power supplied to the motor will be increased and the roller will try to over speed, when compared to the DRAW roller.

However the speed is limited by the stiffness of the substrate and its ability to be stretched.

It is normal to visually monitor the substrate while running and adjust the set point to ensure that the substrate is flat on the process drum.

	Note !
	It is possible that the value for the torque set point is increased to a level that will over speed the roller to a point, where the substrate will slip on the roller and the actual speed will reach the set point. This will be identified by either a “roller slipping” alarm, or by the roller speed indication on the DRIVE HMI screen being much higher than the line “speed”

This mode of operation is described as an open loop system

4.18 Rewind Layarm Roller Assembly

A layarm roller assembly is used to ensure that the substrate is supported until it is wound onto the rewind roll.

The rewind layarm assembly incorporates two rollers mounted on a bracket. This bracket is mounted on a slide rail, which moves in a parallel path as the rewind roll increases in diameter.

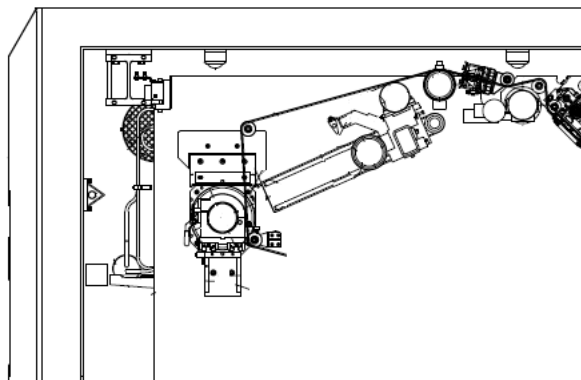


Figure 4.18-1 Rewind layarm arrangement.

There are two modes of operation:

- GAP mode. In Gap mode, a gap is maintained between the last roller before the rewind roll and the rewind roll. The last roller before the rewind roll moves out as the rewind roll increases in diameter. This gap is 25mm as standard.
- Contact mode. In Contact mode, the last roller before the rewind roll presses on to the rewind roll. The roller maintains a fixed pressure on the rewind roll as it increases in diameter.

	Note !
	If excess pressure is applied to the rewind roll, the substrate can be damaged.

Setting the Operating Mode

The mode of operation is set in the Recipes - Winding screen. It can also be changed from the DRIVES HMI screen.

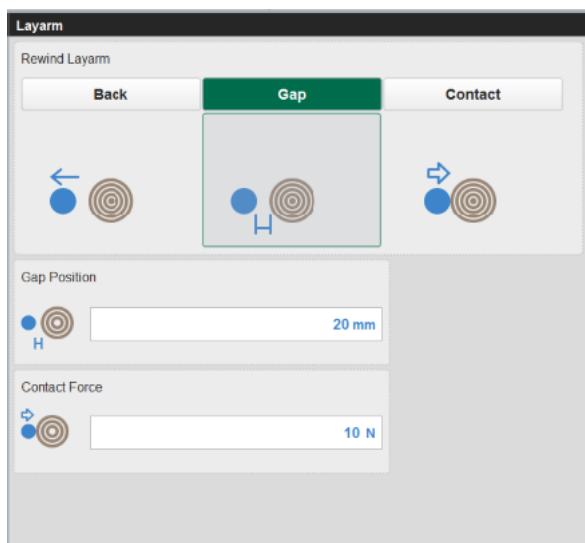


Figure 4.18-2 Recipes - Winding HMI screen.

- Touch the Gap or Contact option to select the mode of operation.
- Touch the Back option to disable the arm from moving, and being used.

The amount of contact pressure applied or gap required is set on DRIVES HMI screen. These values can be changed while the coating cycle takes place, but it is advisable not to change between the modes of operation.

	Note !
	Maximum gap that can be set is 50mm, but it is normal to run at 15-20mm. Maximum contact pressure that can be set is 400 newtons. It is normal to run at 50-100 newtons.

Movement by Operator

The In and Out buttons on the rear control station allow the operator to adjust the layarm assembly. This function is only available with the chamber fully open and winding cart fully retracted.

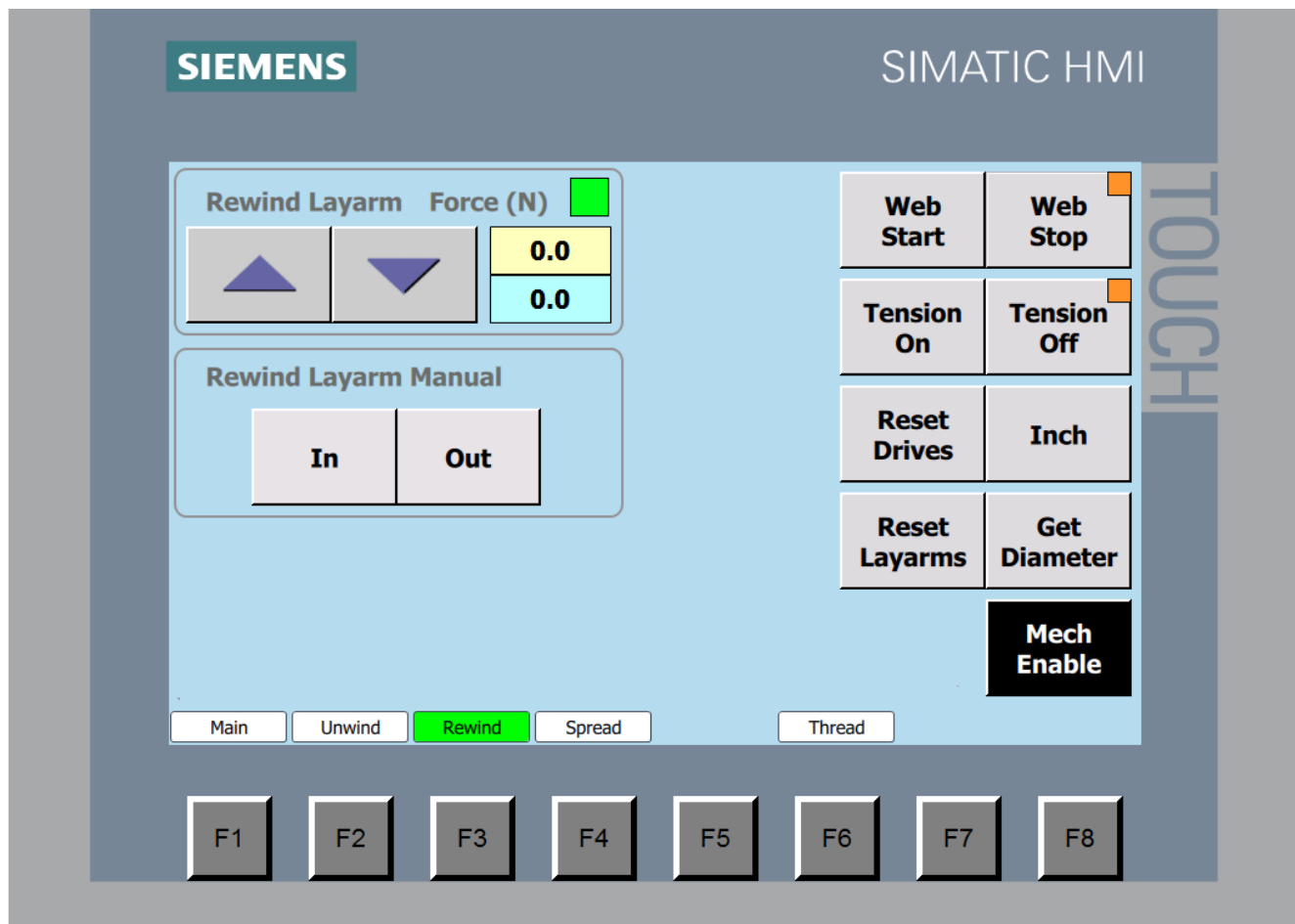


Figure 4.18-3 Rear HMI control station at the loading area.

Limitation of Movement

The upper and lower limits of movement are stopped by mechanical stops at either end of the slide rail.

A proximity switch/sensor also monitors the fully OUT (Home) position.

Modes of Layarm Motors during various periods in operating cycle

At certain times in the operating cycle of the machine, the drive system will override the operator's input in the control of the layarms. These situations are described below; -

- Winding cart open, "Reset Layarms" Command
The layarm will return to the home position, when the home switch is enabled. A message on the drive screen will indicate this condition
- Winding cart open, "Get Diameter" Command

The layarm will move from the “home position” to contact the roll/core and stop further movement. A message on the drive screen will indicate this condition

- Winding cart open, tension applied.

The arm will move from if at the “home” position to the roll/core as part of the tensioning up sequence. It will go to the selected mode of operation, either gap or contact.

- Winding cart open, tension applied, inching of drive system

The layarm will track the roll/core in the mode selected.

- Winding cart inside the chamber, tension applied and the drive interlock activated

The layarm will move to the rewind roll. A timer allows the roller to be driven down. It will then apply pressure to the roll of substrate to stop possible movement of the upper layers.

- Winding cart inside the chamber, tension applied, the drive interlock activated and the line is started

When the line start signal is received, the mode of operation should change (as required) to that selected by the user in either contact or gap mode.

- Winding cart moving out of chamber, tension signal deactivated, drive interlock not activated

As the tension mode is deselected, the layarm assembly will be driven back to its “Home position”. The limit of this travel is stopped by “home” switch inside the chamber.

- Web break situation with winding cart in the chamber.

Layarm will drive into roll at maximum torque level to minimise possible loss of substrate

4.19 Load Cell Rollers

Two sets of carbon fibre lightweight rollers are supplied and are mounted between sets of load cell transducers.

These are normally located in the following areas:-

- Unwind Roll
- Rewind Roll

The rollers rotate on bearings located in the load cell transducers

The purpose of this combination of rollers and transducers is to register the amount of tension being applied to the substrate across its width. This information is then transmitted to the drive system for control and monitoring.

The tension levels are displayed upon the DRIVES HMI screen.

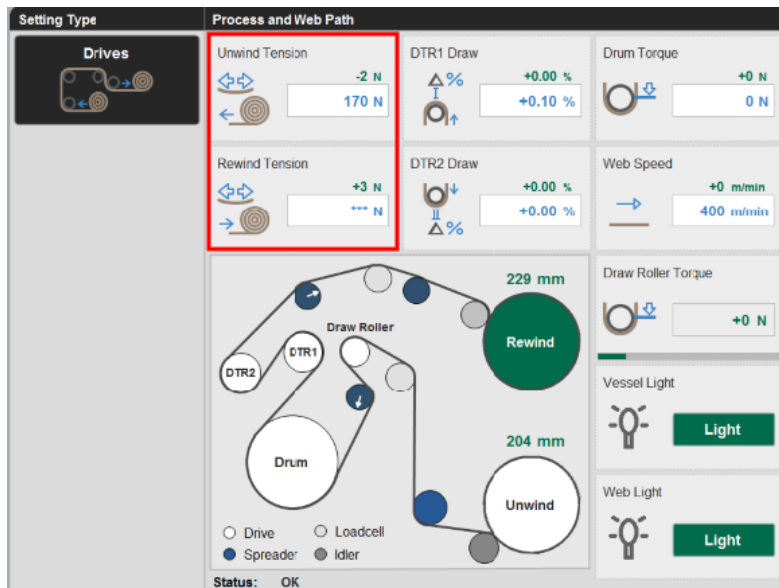


Figure 4.19-1 DRIVES HMI screen highlighting tension feedback.



Figure 4.19-2 PROCESS HMI screen highlighting tension feedback.

4.20 Cut Spreader Roller

A number of these rollers may be located in the machine to spread the substrate. The substrate presses on the roller fingers, which in turn move sideways and spread the film.



Figure 4.20-1 Spreader roller.

The roller's exclusive design is increasingly flexible from the centre to the edges, providing maximum wrinkle removal action without distorting or causing uneven tension throughout the web.

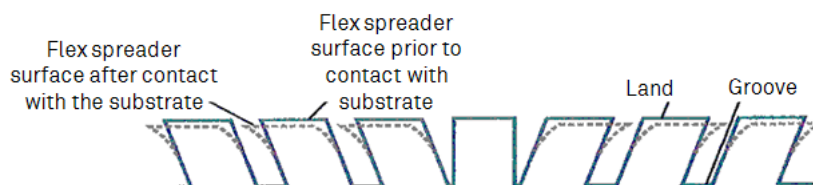


Figure 4.20-2 Straight cut spreader roller principle.

Profile Depth

From the start of the groove, the profile becomes deeper and more angled starting at 6mm and increasing to 30mm at the edges.

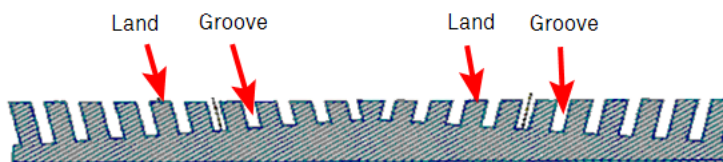


Figure 4.20-3 Variable depth cut spreader roller groove profile.

This roller will become worn over a long period of time. This is noticeable when moving out to a wider width substrate than has been run previously. If there is a problem, it will normally show up as a wrinkling problem.

4.21 Bow Type Spreader Roller

This type of roller is also sometimes called a 'Banana Roller'

A bow type roller may be located in a number of positions within the winding mechanism to spread the substrate. The amount of bow in the roller is fixed.

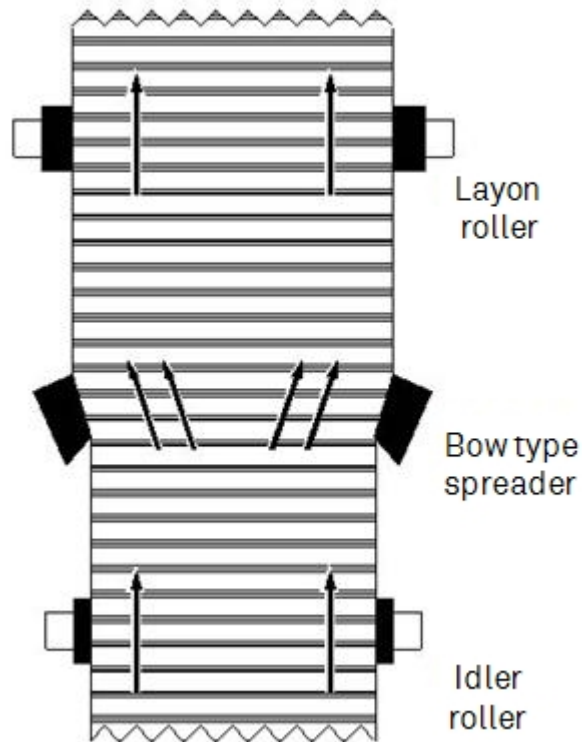


Figure 4.21-1 Operation of bow type spreader roller.

Position of Apex

The apex of the bow can be adjusted at the roller. The bow should be moved depending upon the web path used. Generally, once the test engineer has set it, it will not require any further adjustment. Some machines have a remotely adjustable apex, which can be adjusted whilst the machine is closed.



Figure 4.21-2 Kickert spreader roller apex indicator.

See the vendor manual for information on the operation and positioning of the bow in relationship to the substrate.

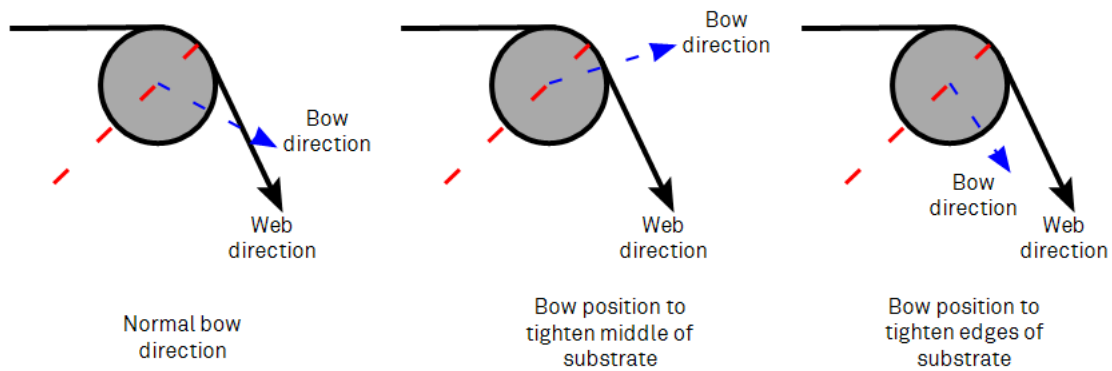


Figure 4.21-3 Bow apex position effects of the spreader roller performance.



Figure 4.21-4 Spreader feedback potentiometer enclosure.

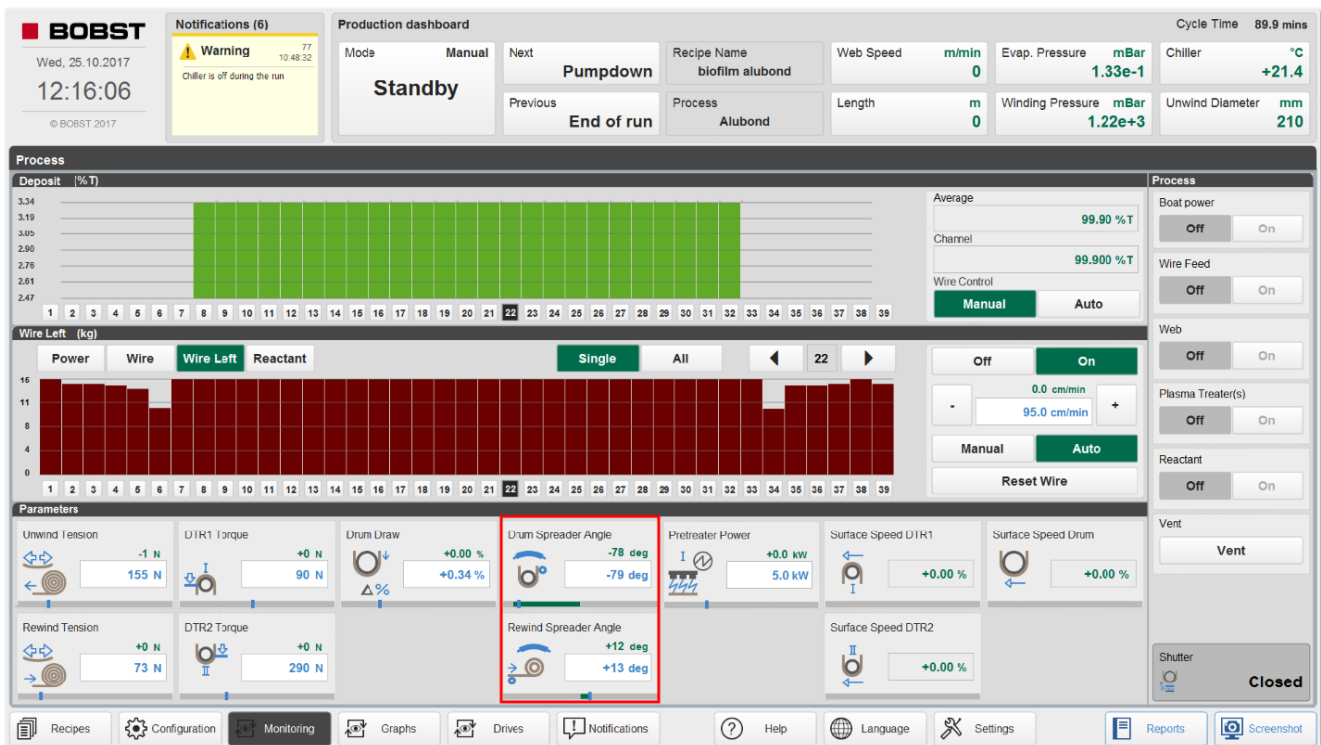


Figure 4.21-5 MONITORING HMI screen indicating spreader roller position set-point and feedback.

4.22 Gas Wedge Assisted Cooling System

A small amount of dry air is placed between the substrate and the process drum to improve the conduction of heat away from the substrate. This provides the following benefits:

- Improved film cooling
- Little or no temperature rise at the rewind roll
- Improved convertibility
- Removes 'tramlines'

Purpose of air injection

In machines where gas is not inserted between the substrate and the cooled process drum, heat transfer from the substrate is only at points where contact are made.

Inserting air in between the substrate and the process drum, provides a medium for the heat transfer to take place.

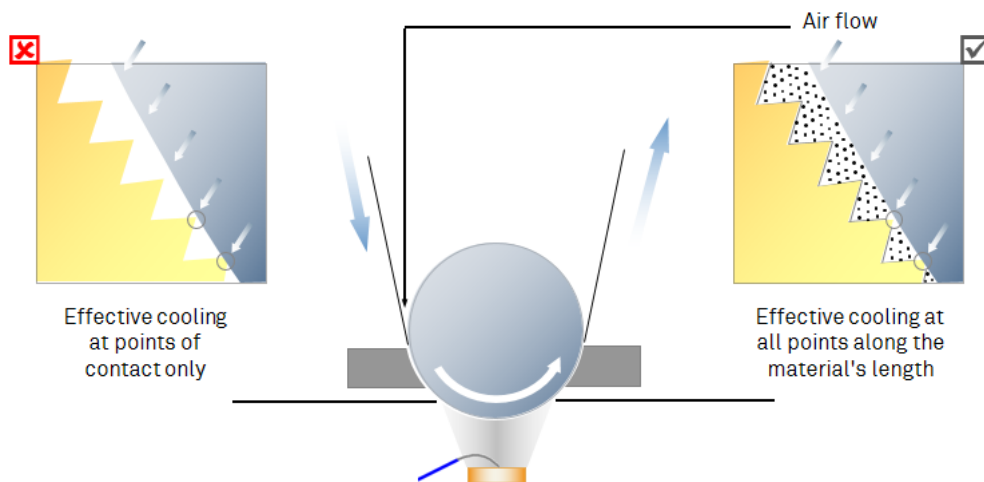


Figure 4.22-1 Drawing explaining effective cooling at points of contact only.

Why is the device, called a Gas Wedge?

The air (gas) is fed into a specially designed distribution device that allows equal gas flow across the full width of the process drum.

Viewing the Gas Wedge location

The gas wedge can be seen from the rear of the machine in location.

There is also a hole in the side of the outboard plate to allow viewing, for checking that the substrate is not touching the assembly.

Adjusting Gas Flow

The amount of air being fed to the system is controlled in the configuration HMI screen.

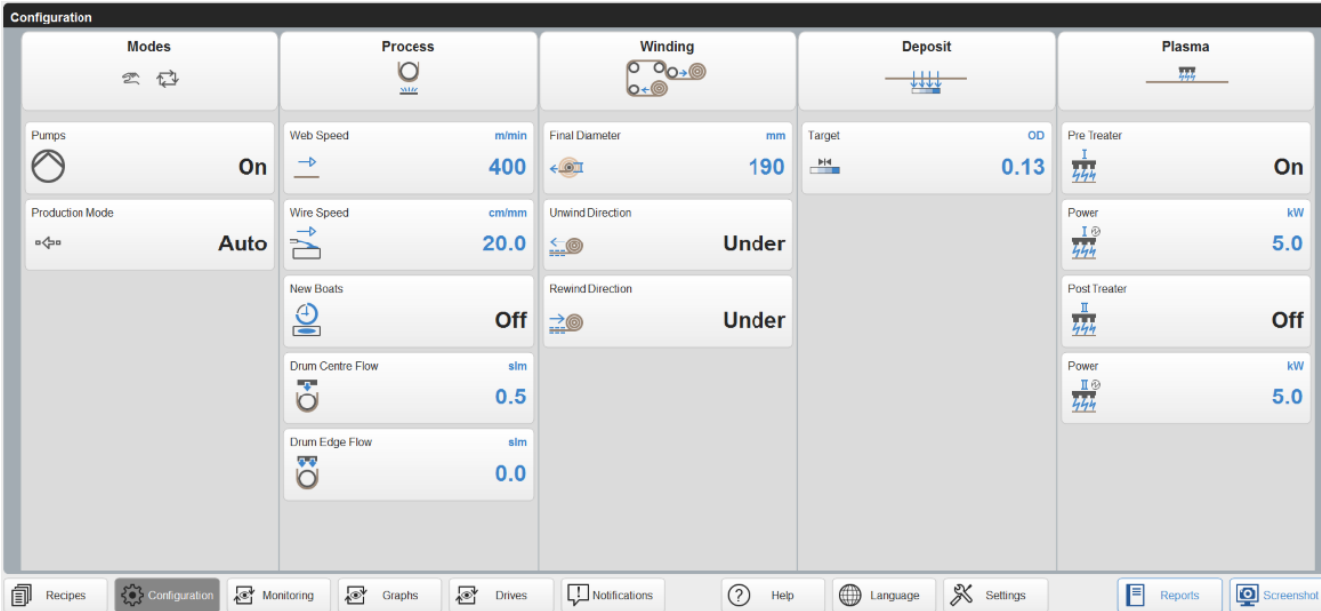


Figure 4.22-2 CONFIGURATION screen showing gas flow controls for gas wedge.

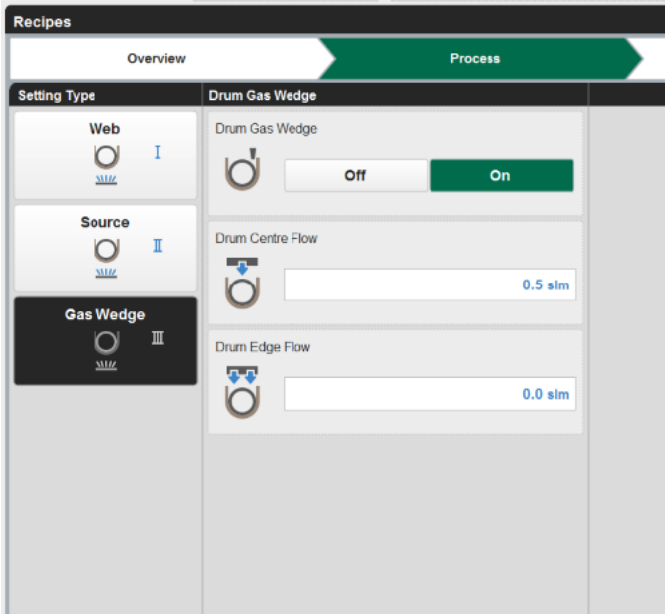


Figure 4.22-3 RECIPES screen capture showing gas flow controls for gas wedge.

	Note!
	If the flow is allowed to be too high, there will be a vacuum breakdown in the evaporation zone. Normal operating level is approximately 4 litres/min in total Excessive gas flow can also affect other aspects of the process, such as the plasma treater performance

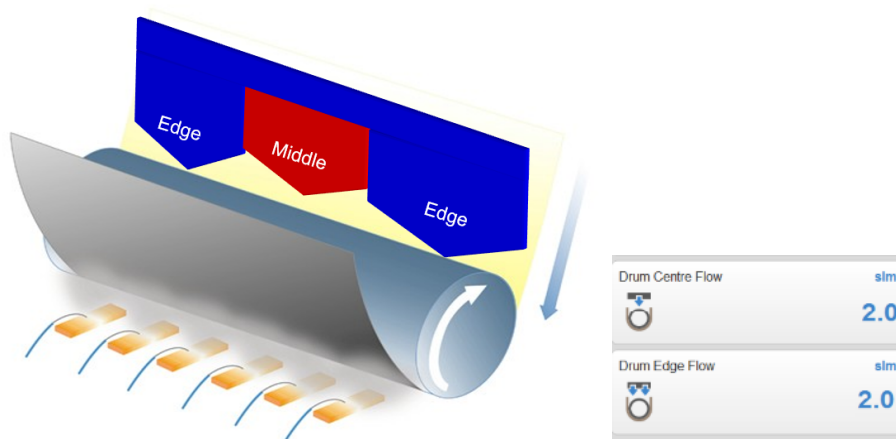


Figure 4.22-4 Drawing showing even gas flow from a 2-zone gas wedge.

The 2-zone gas wedge system allows for changes in the amount of flow as demonstrated in these drawings.

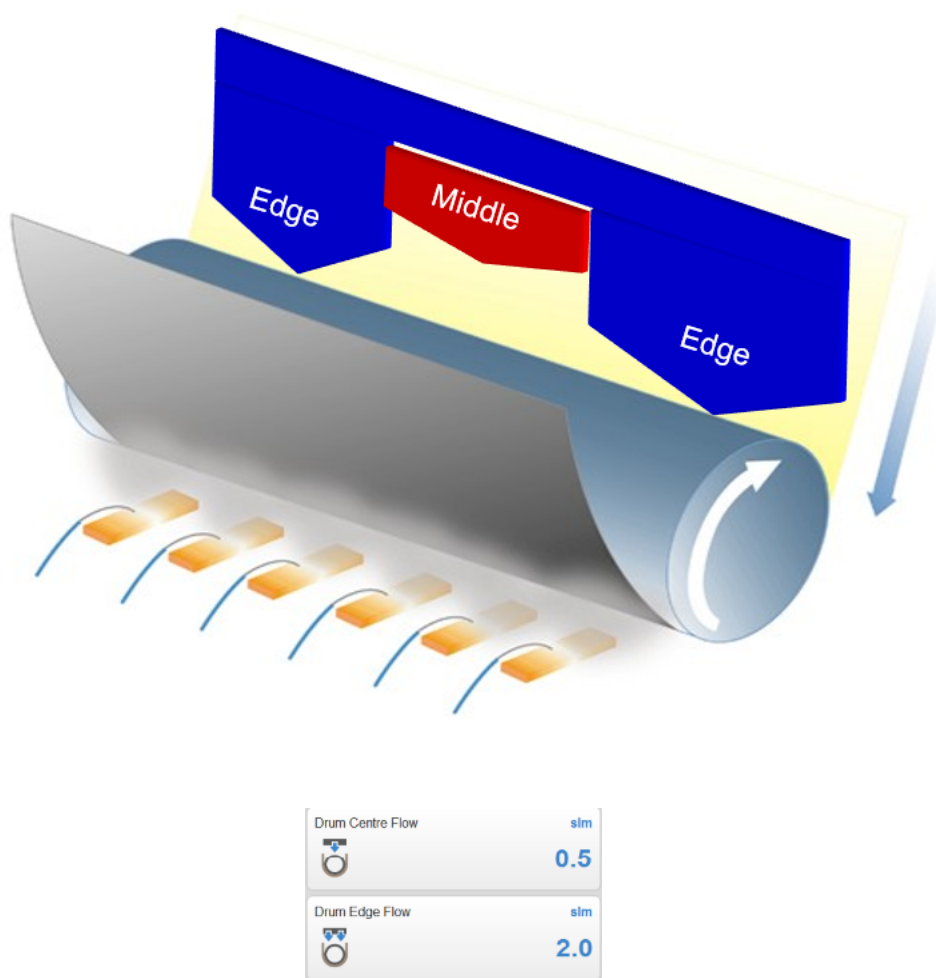


Figure 4.22-5 Drawing showing uneven gas flow from a 2-zone gas wedge.

Removing Tramlines From the Substrate

The word “Tramline” described as phenomena when the substrate is not flat as it passes through the evaporation zone. With heat, a “buckle” forms, and shadowing of the coating takes place.

The tramline can be formed because of a number of reasons

- Expansion of the substrate as the aluminium contacts the surface.
- High co-efficient of friction between the drum surface and the substrate.



Figure 4.22-6 Image of a tramline.

The injection of air between the substrate and the drum allows the substrate to be processed and:

- Reduces expansion by improved film cooling.
- Acts as a 'cushion' to reduce the co-efficient of friction
- Prevents formation of heat wrinkle

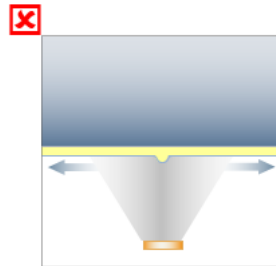


Figure 4.22-7 Image of buckle being formed.

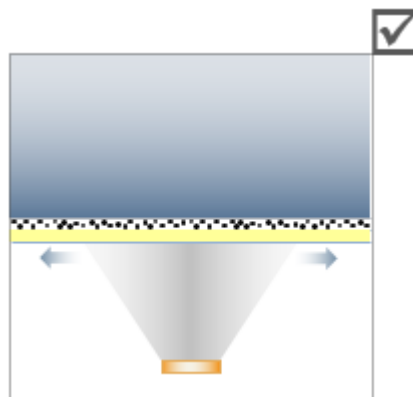


Figure 4.22-8 Image of gas wedge in operation, and no tramline being formed.

It has been found that on thinner materials, for example 20 micron CPP. Excessive gas can actually cause tramlines, so care should be taken when setting the gas flow level.